

# Convergencia puntual dominada implica convergencia en $\mathcal{L}^p$

**Proposición 1.** Sean  $(f_n)_{n \in \mathbb{N}}$  funciones medibles  $f_n: X \rightarrow \mathbb{K}$  en  $(X, \Sigma, \mu)$  dominadas por  $g \in \mathcal{L}^p(X)$  tales que  $f_n(x) \xrightarrow[n \rightarrow \infty]{} f(x)$  c.t.p.  $x \in X$

$$\implies f_n \xrightarrow[n \rightarrow \infty]{\mathcal{L}^p} f.$$

**Demostración:** Tenemos que  $\forall n \in \mathbb{N} : |f_n(x)| \leq g(x)$  c.t.p.  $x \in X$ . Por tanto,

$$\forall n \in \mathbb{N} : |f_n(x) - f(x)|^p \leq (|f_n(x)| + |f(x)|)^p \leq (2g(x))^p \quad \text{c.t.p. } x \in X.$$

Luego, por el Teorema de la Convergencia Dominada, tenemos que

$$\lim_{n \rightarrow \infty} \|f_n - f\|_p^p = \lim_{n \rightarrow \infty} \int_X |f_n(x) - f(x)|^p d\mu(x) \stackrel{\text{TCD}}{=} \int_X \lim_{n \rightarrow \infty} |f_n(x) - f(x)|^p d\mu(x).$$

Como  $\lim_n |f_n(x) - f(x)| = 0$  c.t.p.  $x \in X$ , se tiene que  $\lim_n \|f_n - f\|_p^p = 0$ . ■

### **Temporary page!**

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